AI Collaboration and Role-Evolution Brief
AI Use in the Lead Service Line Infrastructure Analysis

Project Context

I used AI throughout this project as a research and analytical assistant while working with the U.S. EPA's 2026 Q2 Service Line Inventory. The goal was not to let AI make the decisions for me, but to use it to speed up repetitive work, challenge my assumptions, help structure the analysis, and improve how I communicated the results.

The project ultimately analyzed 43,993 complete-report Community Water Systems across the 50 states and District of Columbia. I compared two separate dimensions of infrastructure need: Identification Need, measured by Lead Status Unknown service lines, and Known Replacement Inventory, measured by Lead plus Galvanized Requiring Replacement lines. AI supported the project from data preparation through the final presentation, but I manually reviewed the important methodological choices, calculations, and conclusions.

How AI Supported the Analysis

During data preparation, AI helped me understand the EPA dataset, identify the fields that were relevant to the research question, and write the Python code used to clean and filter the data. It helped me interpret why several numeric columns were stored as text, suggested a safe way to remove commas before converting them to numeric values, and helped organize validation checks around missing values and reported totals.

AI also helped me decide what analysis was actually worth keeping. Early in the process, it suggested additional inspections and diagnostic cells that were technically valid but not necessary for the final notebook. I deliberately removed several of these so that each notebook cell had a clear purpose: move the analysis forward, justify a method, validate an important result, or communicate a necessary finding.

One of the most important uses of AI was in thinking through the population adjustment. The EPA file included a system-level Population Served Count, and at first it seemed convenient to sum that field by state. AI helped raise the concern that system service populations can overlap and may not represent unique residents. After reviewing that issue, I decided not to use the EPA field as the state denominator. Instead, I added 2025 U.S. Census resident population estimates and treated the resulting per-10,000 measures as state-size context rather than estimates of customers affected. This was a case where AI helped surface a methodological risk, but the final decision required human judgment about what the denominator actually meant.

AI also supported the analytical framework itself. It helped separate the problem into Identification Need and Known Replacement Inventory instead of collapsing everything into one score. That distinction became central to the final story because the two measures represent different stages of infrastructure work. Unknown lines still require classification before replacement need can be determined, while known Lead and GRR lines already represent defined replacement work.

What I Verified Manually

I treated important AI suggestions as starting points rather than final answers. I manually checked the row counts after each major filtering step, confirmed that the final analytical sample contained 43,993 systems, and verified that there were no duplicate PWS IDs in the final sample.

I also checked the service-line accounting directly. All 43,993 records reconciled so that Lead + GRR + Unknown + Non-lead matched the reported total service-line count. For the 16 records with missing Non-lead values, I confirmed that Lead + GRR + Unknown already equaled the reported total, so those missing values did not affect the core metrics.

Later in the analysis, I added a direct calculation showing that Identification Need exceeds Known Replacement Inventory in 49 of the 51 states/DC. That calculation was important because I wanted to use the phrase "most states" in the final presentation, and I did not want to make that claim based only on the appearance of a chart.

I also manually reviewed the final notebook and presentation for consistency. This included checking that Unknown lines were never described as confirmed lead lines, that per-capita measures were labeled per 10,000 state residents rather than customers served, and that the numbers in the slides matched the notebook outputs.

Examples of AI Errors or Weak Suggestions

AI was useful, but it also made mistakes and sometimes suggested language or analysis that could have weakened the project if I had accepted it without review.

One example was terminology. At different points, AI suggested phrases such as "confirmed replacement burden." I rejected that wording because "burden" sounded stronger than what the data actually measured, and "confirmed" was unnecessary. "Known Replacement Inventory" was more precise and directly tied to the calculation.

Another example was redundancy. During the notebook work, AI suggested adding another "Population-Adjusted Pattern" section even though one had already been created. I caught the duplication and kept only the original section. This was a useful reminder that AI can lose track of an evolving document even when the individual suggestion sounds reasonable.

AI also sometimes suggested additional analyses simply because they were possible. Examples included more diagnostic checks, more comparisons, and eventually discussion of county-level analysis. The EPA file did not contain a county field, so moving to counties would have required an additional crosswalk and assumptions about systems that serve more than one county. I decided that this would add complexity without improving the main question, so the final project stayed at the state level.

There were also smaller workflow examples. When I first considered using a Census Excel file, AI suggested installing an additional Excel dependency. After I questioned whether that was necessary, we switched to a CSV version instead, which kept the environment simpler. In the final presentation, AI also helped identify clipped labels, crowded scatterplot annotations, and an overly notebook-like slide structure. Those suggestions were useful, but I still made the final choices about which findings deserved emphasis and how the story should be framed.

How AI May Change Junior Analyst Work

This project gave me a practical view of how AI may change entry-level analyst work. AI can reduce the time spent on repetitive tasks such as writing basic pandas code, reviewing field names, drafting validation logic, brainstorming chart structures, and producing first drafts of explanations. It can also act as a useful second set of eyes by raising questions that an analyst may not immediately consider, such as whether a denominator is conceptually valid or whether a chart is communicating the intended message.

At the same time, the project showed me that faster analysis does not remove the need for analyst judgment. In some ways, it makes judgment more important. Someone still has to decide what a metric means, whether a comparison is fair, whether an AI-generated phrase overstates the evidence, which checks are actually necessary, and when an analysis has gone far enough.

I expect AI to shift some junior analyst work away from repetitive syntax and first-draft tasks and toward validation, source interpretation, problem framing, quality control, and communication. The strongest use of AI is not to accept its output automatically, but to use it as a collaborator whose work still needs to be questioned, tested, and refined.